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STATE OF THE ART REVIEW

Digital twins and multimodal artificial intelligence in spine care: a scoping review of concepts, evidence, and translational barriers

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Abstract

Purpose This scoping review examines current evidence supporting multimodal artificial intelligence, continuous monitoring, and digital twin concepts in spine care. Our primary aims were to (1) characterize the state of digital twin development in spine care, (2) identify key technological and conceptual gaps, and (3) evaluate translational barriers to clinical implementation.

Methods A scoping review was conducted following PRISMA-ScR guidelines. PubMed/MEDLINE, Scopus, and Web of Science were searched for studies published between January 2010 and March 2025. Findings were synthesized qualitatively.

Results Twenty-six studies met inclusion criteria. Existing spine prediction models demonstrate modest discrimination and are predominantly static. Imaging-based AI shows weak associations with pain and disability. Wearable sensor monitoring is feasible but lacks consistent evidence for improved outcomes. Spine-specific digital twins remain conceptual, with no prospective validation demonstrating improved decision-making.

Conclusion Multimodal AI-enabled digital twins represent a compelling conceptual framework for personalized spine care, but current evidence does not support clinical superiority or readiness for implementation. Progress will require prospective validation, standardized data integration, and regulatory clarity.

Keywords Spine surgery · Artificial intelligence · Digital twins · Multimodal data integration · Continuous monitoring · Risk prediction

Introduction

Clinical heterogeneity in spine disorders is substantial, as similar structural abnormalities on imaging can be associated with widely divergent pain, disability, and treatment response [1, 2]. Imaging severity correlates only modestly with symptoms and functional limitation, underscoring that radiographic findings alone can be insufficient to guide certain spine interventions [1, 3]. Outcomes are further shaped by comorbidities, psychosocial factors, and prior treatments that may not be captured by conventional assessments [2, 3].

Standard spine care relies on episodic clinic visits and static imaging snapshots that inadequately reflect the nonlinear trajectories of recovery and symptom fluctuation [1, 4]. Widely used risk calculators and single-time-point regression or machine learning models cannot adapt to perioperative events or rehabilitation progress, and their discrimination for complications or patient-reported outcomes remains modest, with typical AUCs in the 0.63–0.80 range [2, 4, 5]. Consequently, prediction is poorly aligned with the lived temporal course of spine disease and recovery [5].

Digital twins have been proposed in light of evolving technology and as a response to the recognition that spinal degeneration, surgical recovery, and nonoperative care evolve dynamically over time [4, 6]. For the purposes of this review, we define digital twins as computational representations of individual patients that are

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continuously updated through bidirectional data exchange between physical and virtual systems, enabling dynamic risk estimation and trajectory forecasting across the care continuum. Conceptually, they aim to provide longitudinal models in which individualized risk estimates and patient states are updated as new clinical, imaging, and functional data accrue, potentially enabling more proactive care [1, 6]. Despite this appeal, spine-specific digital twins remain largely unvalidated, and their incremental benefit over established risk models has not been demonstrated in prospective studies [6, 7]. Figure 1 illustrates the conceptual lifecycle of a spine digital twin, highlighting the integration of multimodal data streams and the proposed longitudinal updating of patient state over time, while distinguishing established components from those that remain investigational.

The primary aims of this scoping review are to (1) characterize the current state of digital twin development in spine care, (2) identify key technological and conceptual gaps limiting clinical applicability, and (3) evaluate translational barriers to implementation. By synthesizing evidence across multimodal AI, continuous monitoring, and digital twin frameworks, we seek to provide a balanced assessment of what this emerging paradigm can and cannot currently offer spine care providers.

Methods

A structured literature identification process was conducted for this scoping review using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for

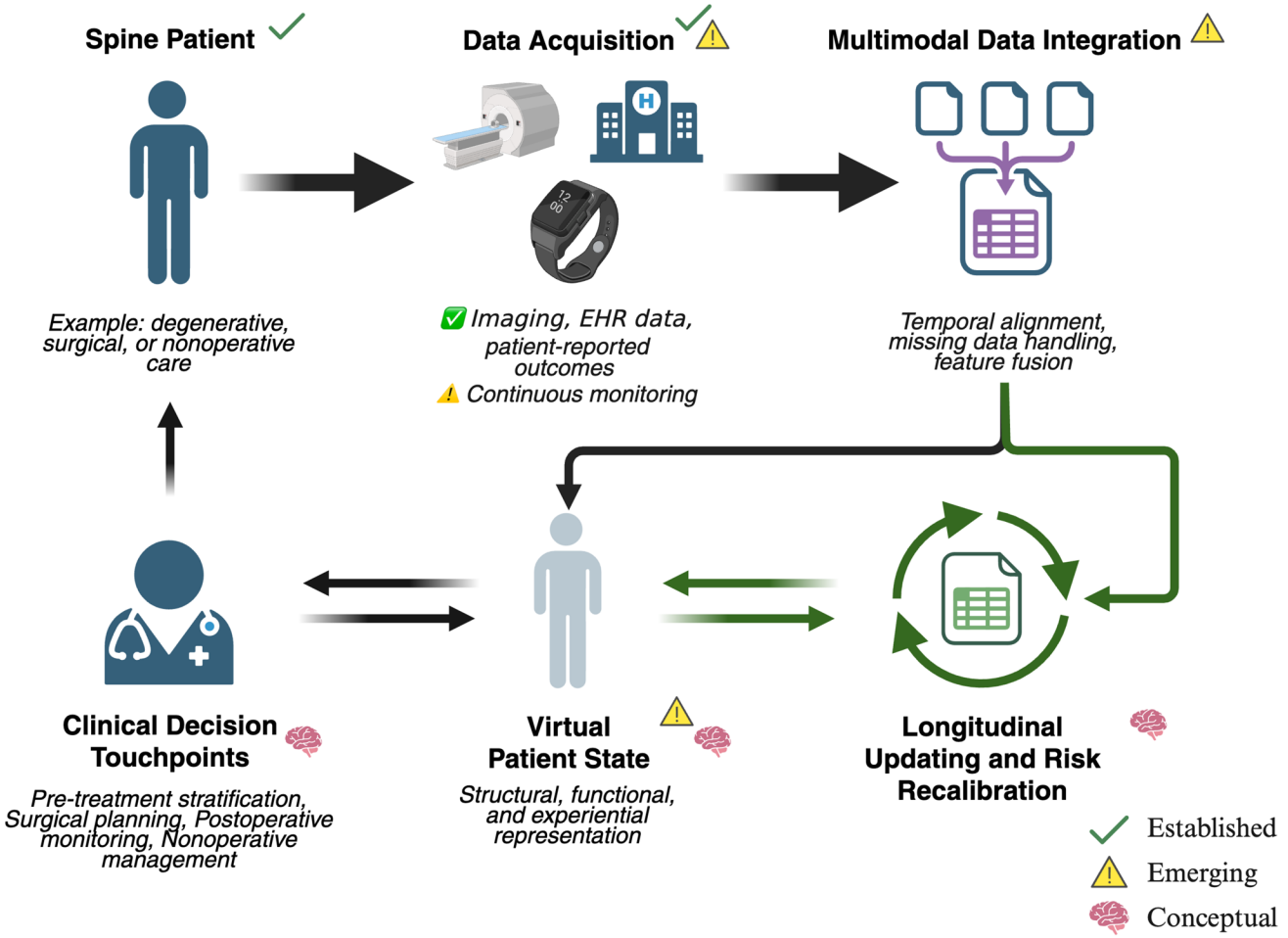


Fig. 1 Conceptual lifecycle of a spine digital twin. Schematic representation of a proposed spine digital twin framework integrating imaging, electronic health record data, patient-reported outcomes, and continuous functional monitoring into a virtual patient state that is updated longitudinally. This figure is included to provide readers with a visual overview of the digital twin concept as applied to spine care, illustrating the proposed data integration architecture and distinguishing currently established components from those requiring further development. Check marks indicate components that are established in current spine care, whereas caution and brain symbols represent capabilities that remain investigational or conceptual. No spine-specific digital twin has yet been prospectively validated for clinical decision-making. Created in BioRender. Kumar, R. (2025) https://BioRender.com/p1l33e3

Scoping Reviews (PRISMA-ScR) guidelines, adapted for conceptual synthesis rather than quantitative meta-analysis (Fig. 2). Figure 2 provides a visual representation of the literature search and screening process, illustrating how studies were identified, filtered, and included based on relevance to multimodal AI and digital twin frameworks in spine care. Two complementary search strategies were employed: one targeting spine-specific applications of artificial intelligence, multimodal prediction, patient-reported and functional outcomes, and continuous monitoring, and a second addressing regulatory, policy, and translational considerations relevant to adaptive and digital twin-based systems. Searches were performed in PubMed/MEDLINE, Scopus, and Web of Science for studies published between January 2010 and March 2025, supplemented by reference list review of key articles. Studies were selected based on relevance to spine care, inclusion of multimodal or longitudinal data elements, and contribution to understanding clinical applicability or translational barriers. Owing to substantial heterogeneity in study design, data modalities, and outcome measures, no meta-analysis or formal risk-of-bias assessment was performed; instead, findings were synthesized qualitatively to contextualize current evidence and identify recurring limitations and gaps.

Multimodal data foundations and the digital twin paradigm in spine care

Results

The literature search identified 982 records (952 from database searching and 30 from additional sources including reference lists of key reviews, seminal conceptual papers, and

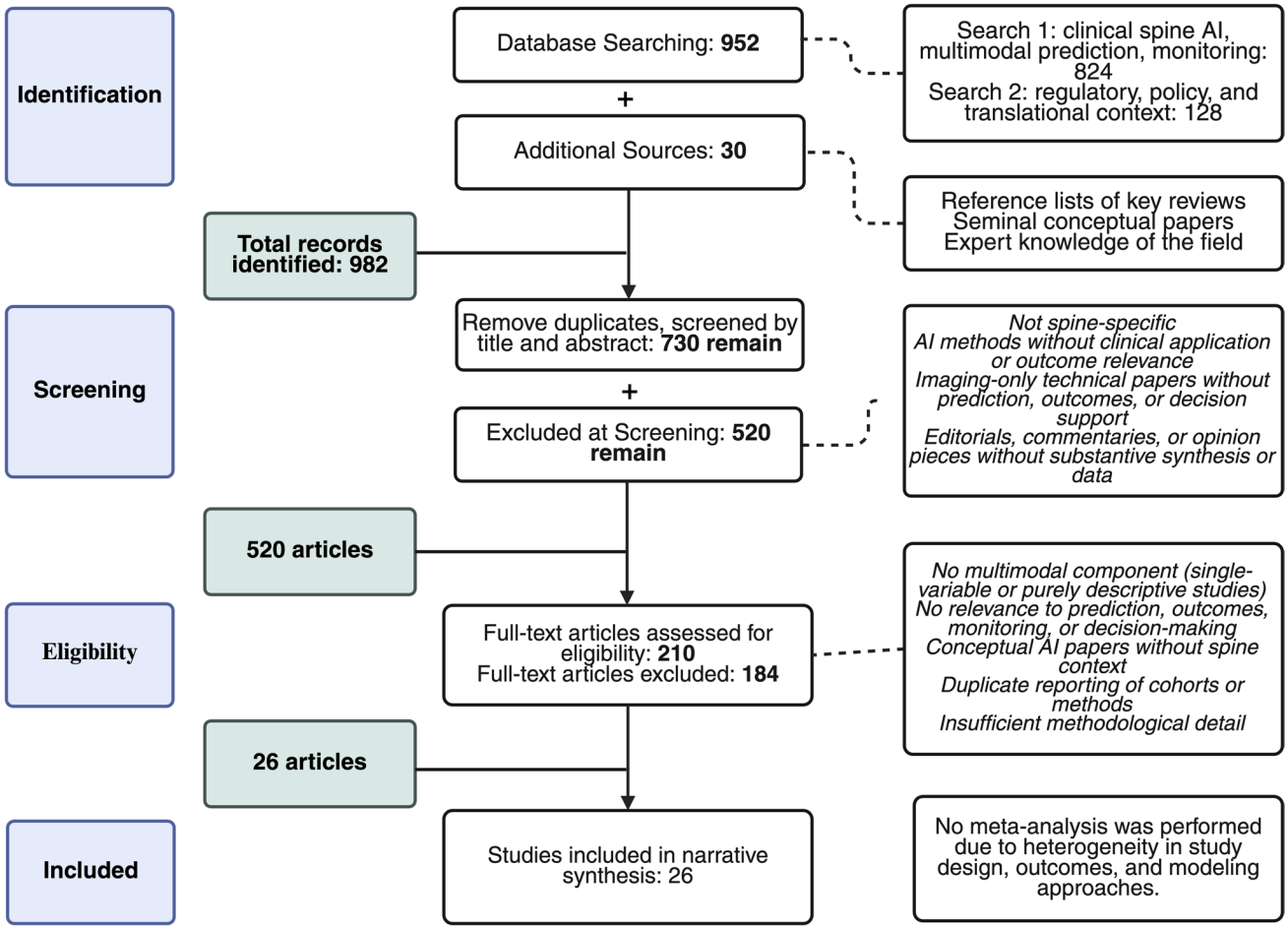


Fig. 2 *Modified* PRISMA-ScR *flow diagram illustrating study identification and selection*. A structured literature search was conducted across clinical, technical, and regulatory domains relevant to multimodal artificial intelligence, continuous monitoring, and digital twin concepts in spine care. This figure is included to provide transparency regarding the literature identification process and to facilitate reproducibility of this scoping review. Study selection followed PRISMA-ScR guidelines adapted for narrative synthesis, emphasizing relevance, methodological contribution, and conceptual insight rather than exhaustive inclusion or quantitative pooling. Created in BioRender. Kumar, R. (2025) https://BioRender.com/b9dim9m

expert knowledge of the field). After removing duplicates, 730 records remained for title and abstract screening. Following screening, 210 full-text articles were assessed for eligibility, of which 184 were excluded. Twenty-six studies met inclusion criteria and were included in the narrative synthesis. Included studies spanned several thematic areas: multimodal AI and prediction modeling (n = 8), imaging-based AI for spine applications (n = 6), wearable and continuous monitoring technologies (n = 5), digital twin conceptual frameworks and prototypes (n = 4), and regulatory and implementation considerations (n = 3). Common terminology across the literature included "multimodal fusion," "continuous monitoring," "longitudinal prediction," and "digital twin," though definitional consistency varied substantially across studies. The discussion that follows is organized into five thematic subsections reflecting these domains: (1) multimodal data foundations and the digital twin paradigm, (2) architecture and longitudinal updating, (3) continuous monitoring and clinical use, (4) comparison to existing models and barriers to translation, and (5) ethical, regulatory, and implementation considerations.

Discussion

Multimodal data foundations and the digital twin paradigm in spine care

Multimodal AI in spine medicine refers to the integration of heterogeneous data sources, including imaging, electronic health record variables, neurologic and functional measures, and patient-reported outcomes, into unified models intended to capture structural, functional, and experiential dimensions of disease [1, 8]. In practice, most spine AI models remain unimodal or bimodal, commonly combining demographic and comorbidity data with limited imaging features, while true high-dimensional multimodal fusion is uncommon [1, 4]. This limitation is consequential because spine outcomes often reflect interactions between anatomy, physiology, behavior, and context that are not represented in any single modality. Table 1 summarizes the primary data modalities relevant to spine digital twins, outlining what each modality captures well, its key limitations, and the current maturity of supporting evidence in spine care.

Imaging-based AI has advanced considerably, with deep learning models achieving high performance for segmentation, alignment measurement, and detection of fractures, stenosis, infection, and tumors [1, 3]. These capabilities provide scalable structural characterization; however, structural metrics exhibit weak and inconsistent associations with pain and disability, and imaging-only models generally underperform those that incorporate clinical and patient-reported data when predicting functional outcomes [1, 4]. Accordingly, imaging is best understood as an essential component of the phenotype rather than a complete representation of disease impact.

Clinical and EHR data, particularly comorbidities, prior surgeries, perioperative factors, and medication use, exert a strong influence on complications and patient-reported outcomes and may outperform imaging alone in many predictive models [2, 5]. However, missingness, coding variability, and institutional practice differences limit the transportability and constrain model performance when applied outside development settings [4, 5]. These constraints underscore that model validity in spine care is shaped as much by data infrastructure and standardization as by algorithm choice.

Objective neurologic and functional assessments such as gait speed, balance, and range of motion are infrequently collected longitudinally outside research settings, perpetuating a gap between structural pathology and measurable function [19, 20]. Sensor-based approaches remain investigational but suggest the potential to detect mobility and balance changes that are not captured in routine clinical documentation [16, 20]. Similarly, patient-reported outcomes remain central to spine care evaluation and trial endpoints, yet are subject to temporal variability and recall bias, and even validated models struggle with missing and inconsistently collected longitudinal PROM data [4, 21]. Together, these patterns highlight a core tension: the measures most aligned with lived experience are often the least consistently captured at scale.

These limitations converge in the challenge of multimodal fusion. Temporal misalignment between data streams, nonrandom missingness, and unequal sampling frequencies complicate integration across modalities, particularly when continuous sensor data must be reconciled with episodic clinic documentation [8, 22]. In the absence of standardized methodological frameworks, published approaches often rely on bespoke architectures, which limits reproducibility and hinders comparison across studies [1, 8]. This methodological fragmentation has direct implications for digital twins, which depend on coherent multimodal integration and longitudinal updating.

Digital twins are computational representations of individual patients that are updated over time through ongoing data exchange between physical and virtual systems [7, 14]. In spine care, a digital twin would theoretically ingest imaging, clinical, and sensor data to maintain an evolving virtual state for risk estimation and trajectory forecasting [6, 7]. Unlike static risk models that assume fixed baseline risk, digital twins are intended to support dynamic recalibration as new data emerge across perioperative and recovery phases [6, 22]. At present, however, no fully validated or clinically deployed spine-specific digital twin exists, and most reported examples remain conceptual or early-stage prototypes [6, 14]. This gap between concept and implementation

Table 1 Data modalities relevant to spine digital twins: strengths, limitations, and evidence maturity. Overview of imaging, clinical, functional, and patient-reported data streams commonly proposed for spine digital twin development. The table highlights complementary strengths and recurring limitations of each modality and emphasizes the uneven maturity of supporting evidence across domains

Data modality	What it captures well	Key limitations	Evidence maturity in spine care	Role in a digital twin
Conventional imaging (radiographs, CT, MRI)	Structural anatomy, deformity, stenosis, and alignment parameters; supports diagnosis, surgical planning, and biomechanical modeling [9, 10]	Weak and inconsistent correlation with pain and disability; protocol and scanner variability; episodic acquisition; limited insight into daily function [1, 9]	Established for diagnosis and planning; AI-based imaging tools are emerging but require further validation [9, 20]	Structural backbone of the twin, defining geometry, alignment, and pathology; inputs for mechanistic and baseline risk layers
Clinical and EHR data (demographics, comorbidities, labs, prior treatments)	Comorbidity burden, perioperative risk factors, medication exposure, and longitudinal care history; strong predictors of complications and PROMs [2, 5]	Missingness, coding variability, and irregular sampling; limited granularity on function and context [4, 5, 11]	Established for traditional risk models; AI/ML applications are emerging in spine surgery [4, 5, 11]	Contextual scaffold for comorbidity-adjusted risk estimation and dynamic recalibration as clinical events occur
Intraoperative data (operative notes, navigation, robotics, physiologic signals)	Surgical approach, treated levels, blood loss, operative time, navigation or robotic trajectories, neuromonitoring trends [12, 13]	Poor standardization and structured capture; often siloed from EHR; underutilized in predictive models [13, 14]	Emerging; early feasibility studies integrating navigation and robotics data [12, 13]	Procedure-specific refinement of the twin around the index intervention, enabling intraoperative context and post-procedural risk updating
Wearable activity and motion sensors (accelerometers, IMUs, smartwatches)	Real-world activity volume, mobility patterns, and sleep metrics; some systems capture gait quality and posture [15, 16]	Signal noise, device heterogeneity, and adherence challenges; lack of consensus on clinically meaningful thresholds; inconsistent correlation with PROMs [16, 17]	Emerging to investigational; feasibility demonstrated, but limited evidence for improved clinical outcomes [16, 17]	Continuous functional layer capturing mobility and activity trends to inform state transitions and early deviation detection
Advanced motion analysis and gait/balance laboratories	High-resolution kinematics, kinetics, and balance metrics; sensitive to subtle functional impairment and compensation patterns [16]	Resource-intensive, lab-based, and rarely collected longitudinally; poor scalability and limited system integration [1, 16]	Investigational; primarily used in research cohorts and small clinical series [16]	High-fidelity functional reference layer for calibration or validation of simpler wearable-derived measures
Neurologic measures (strength, reflexes, sensory testing, EMG)	Focal neurologic deficits, radiculopathy or myelopathy signs, and neural dysfunction [9]	Examiner-dependent and subjective; rarely digitized or stored as structured longitudinal data; limited continuous monitoring options [1]	Established clinically but under-digitized; EMG-based quantification remains investigational	Functional and partially structural insight into neural integrity, informing clinically meaningful state definitions
Patient-reported outcome measures (PROMs: ODI, NDI, VAS, SF-36, PROMIS)	Pain intensity, disability, health-related quality of life, and treatment satisfaction from the patient perspective [18, 19]	Recall and reporting bias; infrequent collection with substantial missingness; overlapping instruments limit comparability [18, 19]	Established as primary endpoints; PROM prediction models are emerging but not widely implemented [11]	Core experiential signal anchoring outcome representation, contextualized alongside objective structural and functional data
Psychosocial and behavioral data (depression, anxiety, sleep, work status)	Psychological distress, coping, expectations, and behavioral factors known to influence pain and recovery [19]	Inconsistently collected; often self-reported or stigmatized; limited standardized integration in spine models [1, 19]	Emerging; increasingly recognized but rarely operationalized at scale in spine AI [11]	Modulatory layer shaping relationships between structure, function, and symptoms to support realistic trajectory modeling
Socioeconomic and environmental data (insurance, occupation, neighborhood, physical demands)	Access to care, physical workload, and environmental constraints affecting recovery and adherence [5]	Sparse and inconsistently captured; often absent from structured EHR; risk of embedding social bias [8]	Emerging in health AI; largely investigational in spine-specific applications [11]	Contextual layer conditioning risk estimates and explaining heterogeneity among patients with similar structural pathology

is important, as most published spine applications still function as one-time predictions rather than adaptive systems capable of real-time learning [1, 5].

Digital twin implementations can be broadly grouped into mechanistic, data-driven, and hybrid approaches. Mechanistic twins based on finite element analysis offer biomechanical interpretability but require detailed anatomical modeling and substantial computational resources [12, 13]. Data-driven approaches scale more readily but often lack physiologic grounding and are perceived as opaque [1, 8]. Hybrid architectures that combine mechanistic and learned features are emerging but remain experimental [6, 13]. Across these approaches, temporal modeling is central because spinal degeneration and recovery follow heterogeneous, non-linear trajectories, yet existing prediction models rarely encode explicit disease or recovery states and often fail to capture clinically relevant transitions over time [5, 14, 21, 22].

Despite meaningful progress across individual data streams and modeling paradigms, a consistent set of limitations continues to impede the realization of fully integrated, clinically actionable spine digital twins. The principal barriers in this domain include the weak correlation between structural imaging findings and clinical outcomes, the absence of standardized multimodal fusion methodologies, and the limited availability of longitudinal data capturing the temporal dynamics of spine disease and recovery. True high-dimensional multimodal integration remains uncommon, and no spine-specific digital twin has been prospectively validated.

Architecture and longitudinal updating of spine digital twins

Anatomical and biomechanical components typically begin with patient-specific 3D reconstructions derived from CT or MRI, followed by meshing and assignment of material properties [12, 13]. While segment-level modeling may estimate stress distributions relevant to implant failure or junctional kyphosis, increased granularity limits scalability for routine clinical use [13, 14]. As a result, simplifying assumptions are common, creating a persistent tension between physiologic fidelity and feasibility [13].

Imaging-derived features are increasingly extracted automatically, enabling high-throughput quantification of alignment and degenerative parameters [1, 3]. However, domain shift arising from scanner variability and site-specific preprocessing can degrade performance outside development settings, and structural changes often lag behind symptom evolution, limiting imaging as a timely surrogate of patient state [1, 4]. This further reinforces the need to integrate non-imaging signals when the goal is dynamic patient-state estimation.

Longitudinal state representation relies on time-series and latent state-space models that summarize complex histories into compact dynamical representations [22, 23]. Yet there is no consensus on clinically meaningful spine states, and proposed state-transition frameworks lack prospective validation [21, 23]. In parallel, online model updating introduces risks of instability and concept drift when driven by noisy or biased data, a concern that is heightened when data streams are sparse, incomplete, or unevenly sampled [22, 23]. Few spine AI studies report uncertainty estimates or define actionable thresholds that link probabilistic outputs to clinical decisions, limiting interpretability and clinical utility [1, 21].

Taken together, these architectural considerations highlight that current spine digital twin frameworks are constrained by trade-offs between physiologic fidelity, scalability, and robustness. Biomechanical models remain computationally intensive and sensitive to simplifying assumptions, while imaging-derived features are vulnerable to domain shift and often lag behind clinical state changes. Longitudinal modeling approaches lack consensus definitions of clinically meaningful spine states, and online updating introduces risks of instability when driven by sparse, noisy, or biased data streams. Across approaches, uncertainty quantification and clinically actionable decision thresholds are infrequently reported, limiting interpretability and impeding translation into routine spine care.

Continuous monitoring and clinical use across the spine care continuum

Wearable and motion-sensing technologies enable longitudinal capture of real-world activity, mobility, and sleep patterns [16, 20]. These tools can track postoperative recovery and identify deviations from expected activity trajectories, but signal noise, device heterogeneity, and adherence challenges remain significant [16, 17]. Simple metrics such as step count demonstrate inconsistent correlation with clinical outcomes, emphasizing that meaningful clinical translation requires contextualized features and validated analytic pipelines [20]. To clarify how digital twins differ conceptually from existing approaches, Table 2 contrasts traditional spine risk models, single- or multimodal AI models, and the digital twin paradigm across key dimensions including temporal updating, personalization, and validation status.

Mobility and posture tracking provide insight into day-to-day function, with sedentary behavior and reduced gait speed associated with poorer outcomes after lumbar surgery [16, 17]. Interpretation is limited by the absence of standardized pipelines and by contextual factors such as occupation and environment that influence activity independent of

Table 2 Conceptual comparison of traditional spine risk models, AI-based prediction models, and digital twins

Feature	Traditional risk models	Single- or multimodal AI models	Digital twin paradigm
Temporal updating	One-time or episodic predictions derived from baseline or perioperative variables; risk estimates remain static once calculated [5, 11]	Typically trained on cross-sectional or limited longitudinal data; some models ingest time-stamped features, but continuous online updating is uncommon in routine use [1]	Conceptually designed for continuous or frequent updating as new imaging, EHR, and sensor data accrue, allowing recalibration of risk and patient state over time [7, 14]
Individualized trajectories	Represent average risk from population cohorts; can stratify individuals but do not model personalized recovery or degeneration trajectories [5]	May capture non-linear relationships and limited temporal patterns, but typically produce point estimates or short-horizon forecasts rather than full patient-specific trajectories [1]	Aim to maintain an evolving virtual representation encoding individualized structural, functional, and symptom trajectories, enabling exploration of alternative future scenarios [14, 25]
Use of continuous monitoring	Do not incorporate continuous data; rely on static clinical variables and snapshot imaging or PROMs [5]	Rarely integrate high-frequency wearable or home-monitoring data; when included, signals are often aggregated, limiting temporal resolution [17]	Envisioned to ingest continuous or near-real-time signals such as activity, gait, ePROMs, and intraoperative feeds, updating the virtual state in response to observed patient behavior and physiology [7, 14]
Multimodal integration	Typically limited to structured clinical and demographic variables, sometimes with a small number of imaging-derived features; minimal fusion of functional or contextual data [11]	Can integrate multiple modalities, including EHR, imaging, and PROMs, but many spine applications remain single-modality and use ad hoc fusion strategies [1, 25]	Defined by integration of heterogeneous data streams such as imaging, biomechanics, EHR data, wearables, and PROMs into a unified, temporally coherent patient representation [7, 14]
Interpretability	Generally transparent, based on regression or scoring systems with clinically familiar inputs; may oversimplify complex relationships [5]	Highly variable; simpler or tree-based models may be partly interpretable, whereas deep learning models are often opaque and rely on post hoc explanation methods [25]	Conceptually able to embed mechanistic or biomechanical components that enhance interpretability, but full multimodal implementations are likely complex and partially opaque without dedicated explanation layers [7, 14]
Clinical validation status	Several models have undergone internal and limited external validation; discrimination is typically modest and impact studies are rare [5]	Numerous retrospective internal validations; fewer robust external validations, and almost no randomized trials demonstrating improved decision-making or outcomes in spine care [11, 24]	Largely absent in spine care; current work is conceptual or early proof-of-concept with no prospective or randomized evidence of clinical benefit [7, 14]
Regulatory readiness	Often used as informal calculators or embedded tools; rarely classified as regulated medical devices [5]	Some AI models are entering regulatory pathways as software as a medical device, but most spine-specific tools remain investigational [25]	Adaptive, continuously learning systems challenge existing regulatory frameworks; guidance is emerging but remains high-level and incomplete [14, 26]
Implementation in routine spine care	Widely accessible and easily applied, though uptake varies and use is often informal [11]	Limited routine deployment; barriers include EHR interoperability, workflow integration, and clinician trust, confining most applications to research or pilot settings [1, 25]	Currently conceptual and research-focused; requires substantial data infrastructure, interoperability, and governance, with near-term use likely restricted to pilot or simulation environments [7, 14]

Comparison across key attributes, including temporal behavior, personalization, data requirements, interpretability, and clinical validation. Digital twins are distinguished primarily by their proposed capacity for longitudinal updating rather than by demonstrated superiority in predictive performance or clinical outcomes

recovery [1, 20]. High-frequency symptom reporting similarly reveals temporal variability in pain and disability that is missed by periodic clinic assessments [19, 20]. However, increased reporting frequency raises patient burden and contributes to missing data, even though symptom trajectories remain indispensable for clinical decision-making [4, 19].

These monitoring modalities are often framed as enabling a feedback loop in which risk estimates are recalibrated and targeted interventions are triggered as new data accrue [6, 7]. In spine care, however, closed-loop implementations remain unproven, with no randomized trials demonstrating improved outcomes, reduced complications, or lower costs compared with standard care [17, 21]. This limitation is consequential for proposed digital twin use cases, which frequently assume that real-time monitoring can be translated into actionable clinical decisions.

Across the care continuum, digital twins have been proposed as tools for pre-treatment stratification through simulation of alternative strategies, yet evidence that such approaches improve patient selection or outcomes is lacking [5, 21]. In surgical planning, prototypes suggest feasibility for integrating anatomy, biomechanics, and probabilistic risk modeling, but benefits for complication rates or long-term outcomes remain unproven [7, 12]. Postoperative monitoring using wearables and ePROMs may identify outlier recovery trajectories, but routine implementation has not consistently reduced readmissions or complications [4, 17]. In nonoperative care, wearable-guided rehabilitation has shown mixed results and no clear advantage over conventional therapy [16, 20]. For chronic degenerative disease, longitudinal digital twin models are constrained by limited availability of dense, long-term multimodal data in routine practice [1, 23].

Collectively, these findings indicate that while continuous monitoring technologies are feasible and conceptually aligned with the digital twin paradigm, their clinical impact in spine care remains unproven. Wearable-derived metrics show inconsistent associations with patient-centered outcomes, device heterogeneity and variable adherence complicate standardization, and analytic pipelines lack validation across diverse care settings. Importantly, no randomized trials have demonstrated that continuous monitoring improves outcomes, reduces complications, or lowers costs compared with standard care, and closed-loop implementations that translate real-time data into actionable clinical decisions remain largely theoretical.

Comparison to existing spine risk models and barriers to translation

Traditional spine risk models are static and population-derived, with modest discrimination and frequent deficiencies in external validation and impact assessment [5, 21]. Single-modality machine learning models show variable performance relative to regression and share similar temporal limitations [5, 23]. Digital twins are hypothesized to address these limitations through continuous, individualized risk estimation [6, 7]. However, direct comparisons with established models are lacking, and there is no evidence demonstrating superior predictive performance or clinical benefit in spine care [5, 14].

Translation is further constrained by data heterogeneity, imaging variability, and limited external validation, which undermine robustness and transportability [1, 5, 21]. Poor interoperability among EHRs, imaging systems, and wearable platforms limits real-time integration and increases clinician burden, while trust in opaque AI tools remains low in the context of evolving regulatory frameworks [8, 14]. These challenges indicate that the primary bottlenecks are often infrastructural and procedural rather than purely technical.

In current practice, digital twins have not yet demonstrated clear advantages over existing spine risk models. Evidence directly comparing their performance to established approaches is lacking, and real-world implementation is limited by data heterogeneity, poor interoperability across clinical systems, and unresolved workflow and regulatory challenges. As a result, digital twins in spine care remain largely conceptual, with clinical adoption constrained more by practical barriers than by algorithmic capability.

Ethical, regulatory, and implementation considerations

Explainability is central to clinician trust and informed consent, yet efforts to improve transparency can add complexity to already sophisticated systems [8, 21]. Continuous monitoring raises privacy and data ownership concerns, and regulation of adaptive AI remains in flux, with emerging requirements for change control and post-market surveillance [24, 26]. Accountability for AI-informed decisions and sustaining patient engagement over time remain unresolved challenges, particularly when disengagement is patterned and may exacerbate inequities [4, 17].

Collectively, these challenges highlight persistent gaps in governance and implementation. Explainability requirements may conflict with model complexity, regulatory frameworks for adaptive AI in healthcare continue to evolve, and standardized solutions for privacy in continuous monitoring remain lacking. Clear accountability structures for AI-assisted clinical decisions have yet to be established, and patterned patient disengagement may introduce or exacerbate health disparities.

Conclusion and future directions

Prospective validation is essential to establish clinical credibility of spine digital twins, as no randomized trials have demonstrated benefit to date [5, 21]. A randomized controlled trial evaluating digital twins in spine care might enroll patients undergoing elective lumbar fusion, randomized to standard preoperative evaluation versus digital twin-guided decision support. The intervention arm would receive individualized risk estimates and simulated outcome trajectories integrated into shared decision-making consultations. Primary endpoints could include patient-reported outcomes (ODI, VAS) at 12 months, with secondary endpoints addressing complication rates, reoperation, and decision regret. Feasibility considerations would include infrastructure for multimodal data integration, clinician training, and patient consent for continuous monitoring. Such trials would require multi-site collaboration to ensure adequate power and generalizability. Integration with surgical technologies and closed-loop systems remains aspirational, and near-term impact is likely incremental and research-focused rather than transformative [1, 14]. Translational pathways will likely proceed through several phases: initial validation studies comparing digital twin predictions with observed outcomes, followed by implementation research examining workflow integration and clinician adoption, and ultimately pragmatic trials assessing patient-centered outcomes in routine practice. Standardization of data elements, interoperability frameworks, and regulatory guidance for adaptive AI systems will be prerequisites for broader adoption. Multimodal AI-enabled digital twins remain a compelling conceptual scaffold for longitudinal spine care, but their maturation will depend on validation, standardization, interoperable data infrastructure, and implementation pathways that support safe and clinically meaningful use [6, 7].

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Declarations

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